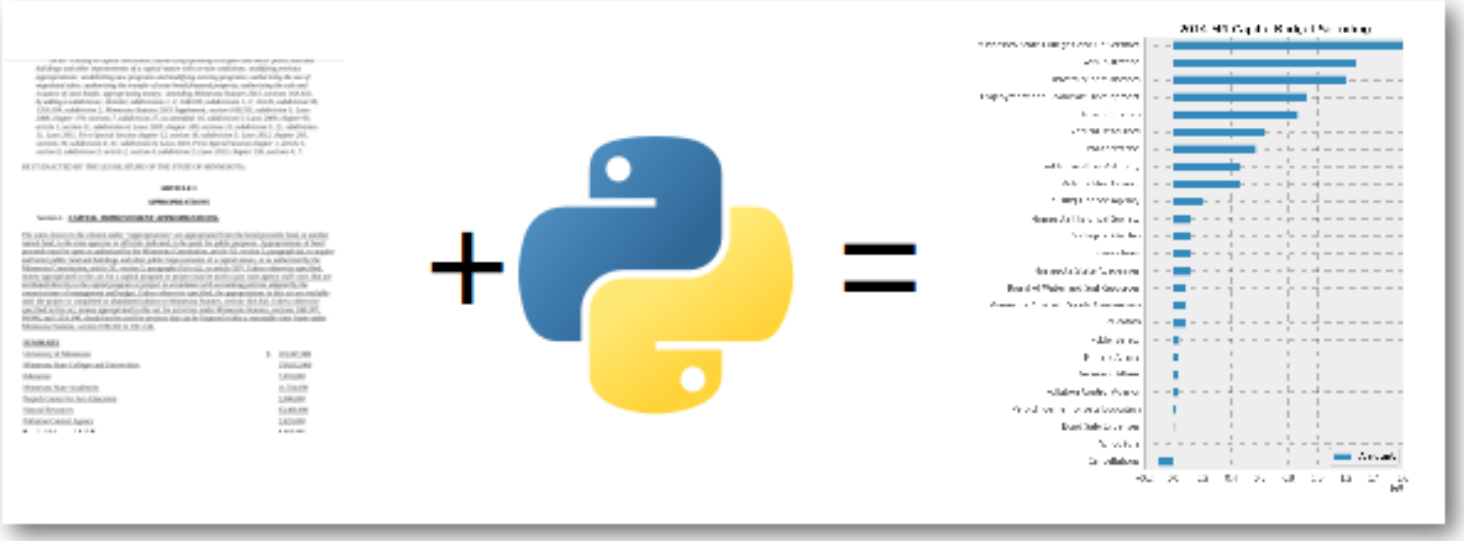
Lyrics Web Scraping and Text Mining Analysis



ECE 143: Group 2

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Introduction

- Goal:

  To study top 100 songs on billboard year-end charts from year 1959 to 2018

- Dataset:

  1. Wiki – Billboard year-end 100: https://en.wikipedia.org/wiki/Billboard_Year-End
  2. Years - 1959-2018
  3. Number of songs - 60x100 = 6000

- Methodology:

  1. Extract data from various websites
  2. Choose relevant variables, such as artist nationality, lyrics, genre, etc.
  3. Perform Data Cleaning, Analysis and Text Mining

Web Scraping - 4 main components

- Part I: Rank, Song, Artist - obtained from Wikipedia
- Part II: Nationality - obtained from Wikipedia
- Part III: Genres - obtained from DBpedia resources
- Part IV: Lyrics - obtained from the Genius database

Web Scraping

rank	song	artist	year	nationality	genres
1	The Battle of New Orleans	Johnny Horton	1959	United States	'Country_music', 'Parody_music'
2	Mack the Knife	Bobby Darin	1959	U.S.	'Jazz', 'Jazz_music', 'Pop_music'
3	Personality	Lloyd Price	1959	U.S.	'Popular_music', 'Rhythm_and_blues'
4	Venus	Frankie Avalon	1959	U.S.	'Pop_music'
5	Lonely Boy	Paul Anka	1959	Canada	'Pop'
6	Dream Lover	Bobby Darin	1959	U.S.	'Rock_and_roll'
7	The Three Bells	The Browns	1959	United States	'Contemporary_folk_music', 'Countrypolitan'
8	Come Softly to Me	The Fleetwoods	1959	United States	-
9	Kansas City	Wilbert Harrison	1959	North Carolina US	'Rhythm_and_blues', 'Rock_and_roll'
10	Mr. Blue	The Fleetwoods	1959	United States	'Popular_song'

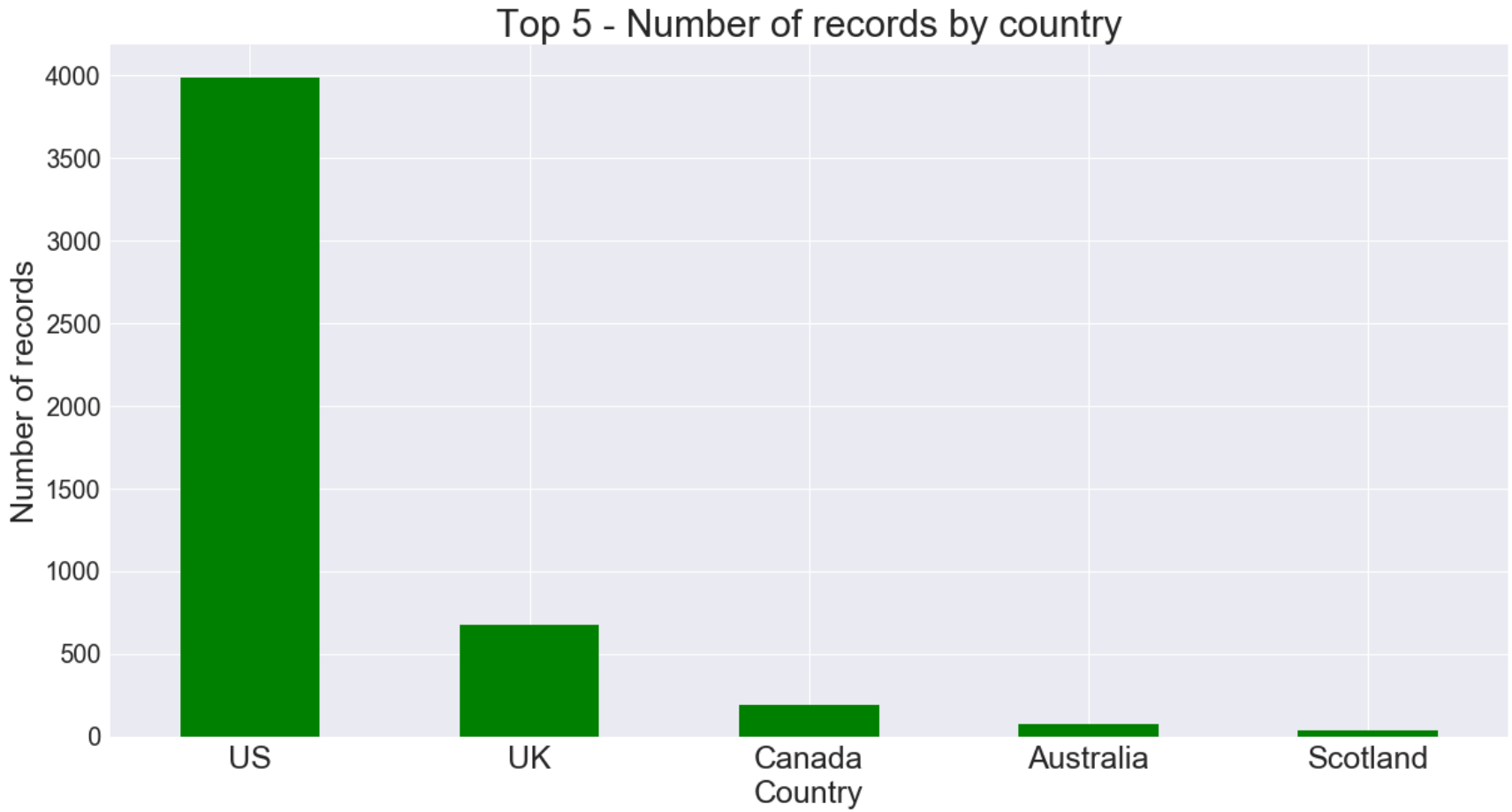
Data Cleaning Needed!

Data Cleaning

- Nationality: Total 128 different nationalities listed by wiki - categorized into 37 nationalities
- Genre: Total 489 genres listed by DBpedia - categorized into 17 main genre classes
- Lyrics: Removal of periods, punctuations, incomplete words

Number of songs

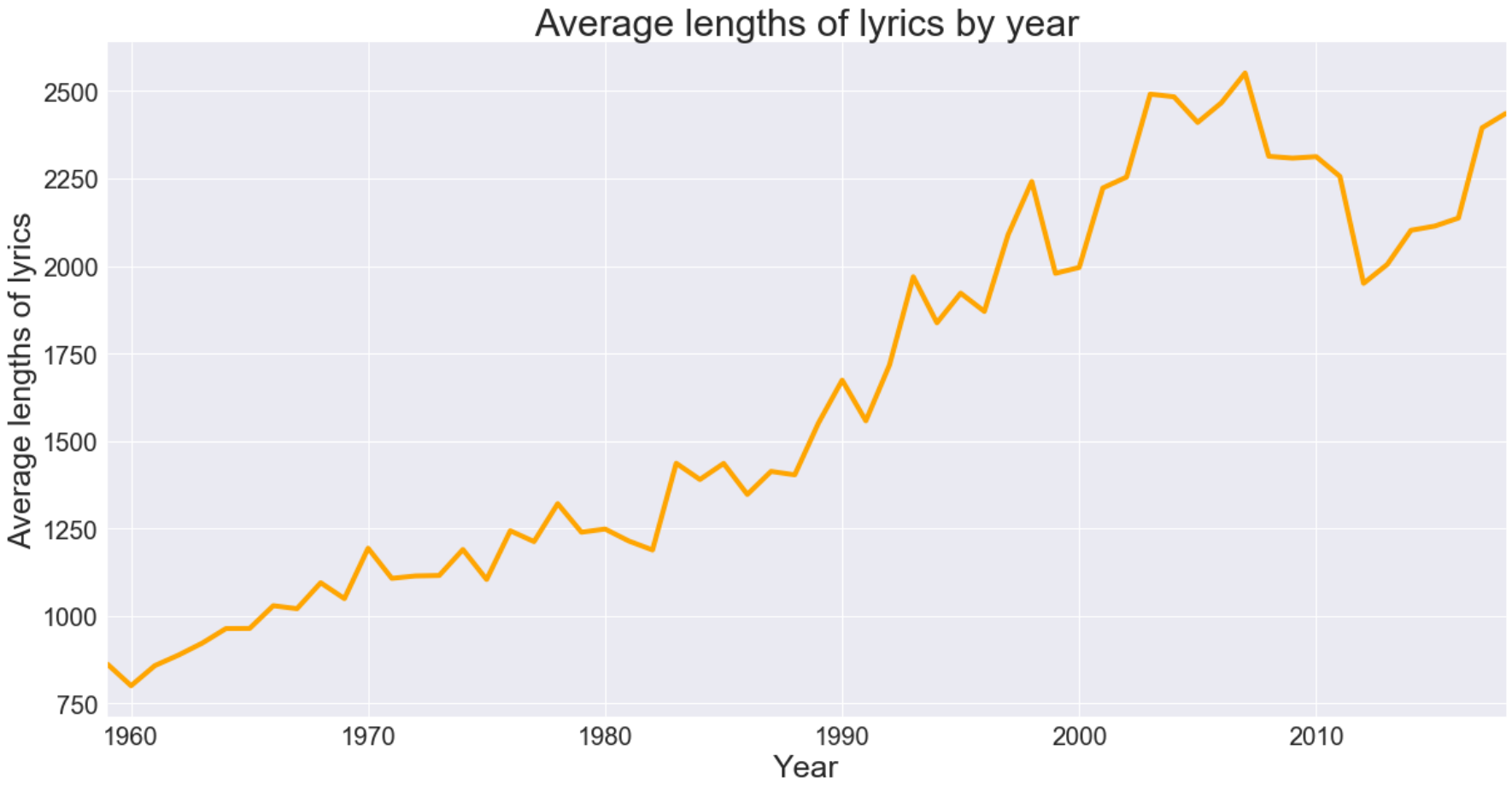
➢ By Country:



US tops the chart!

Average length of lyrics

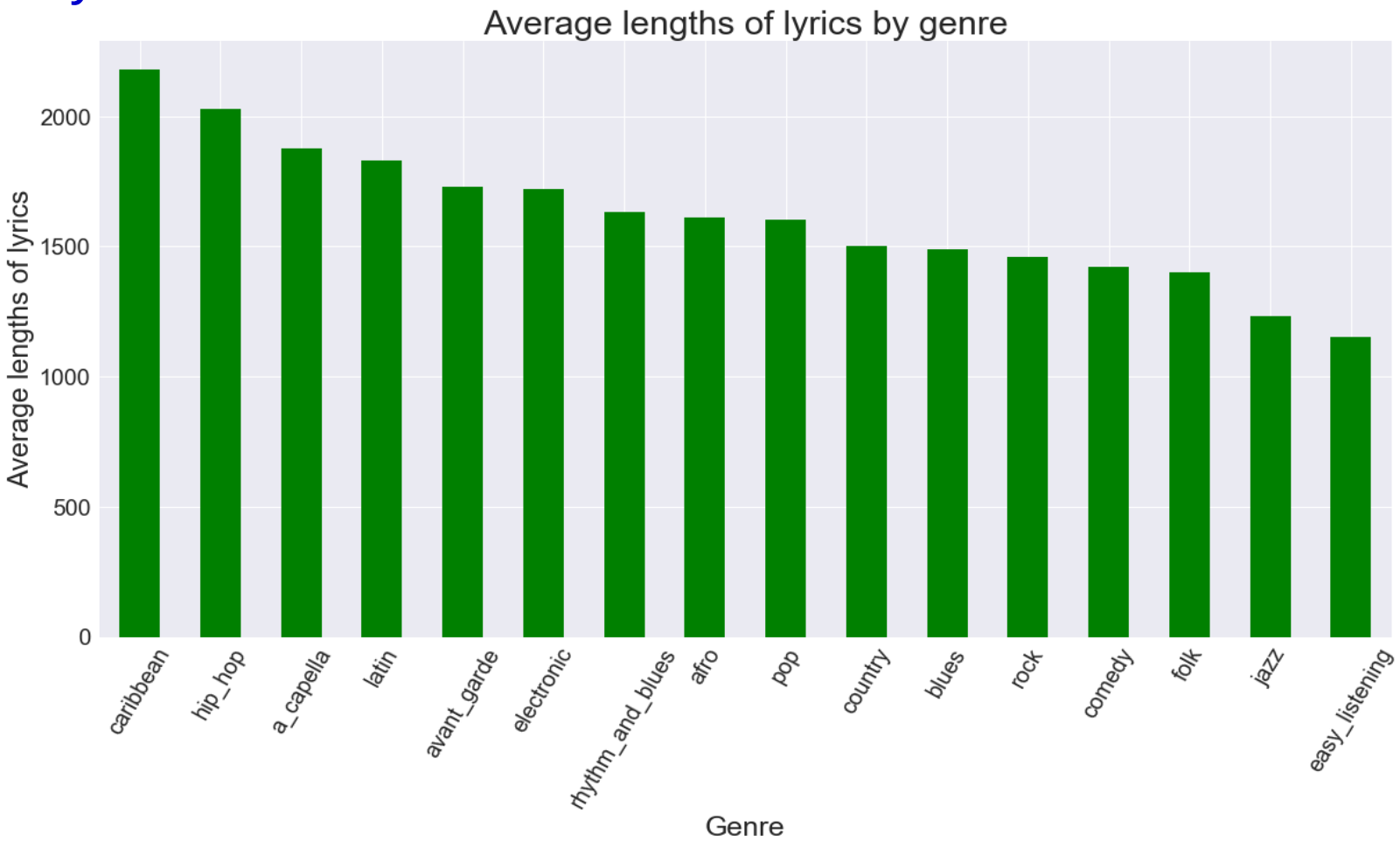
➢ By Year:



Increasing trend!

Average length of lyrics

➢ By Genre:

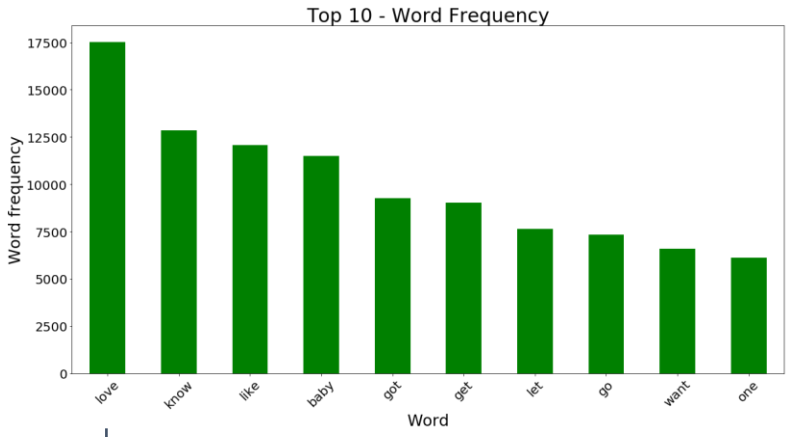


Caribbean music are longest!

Text Mining

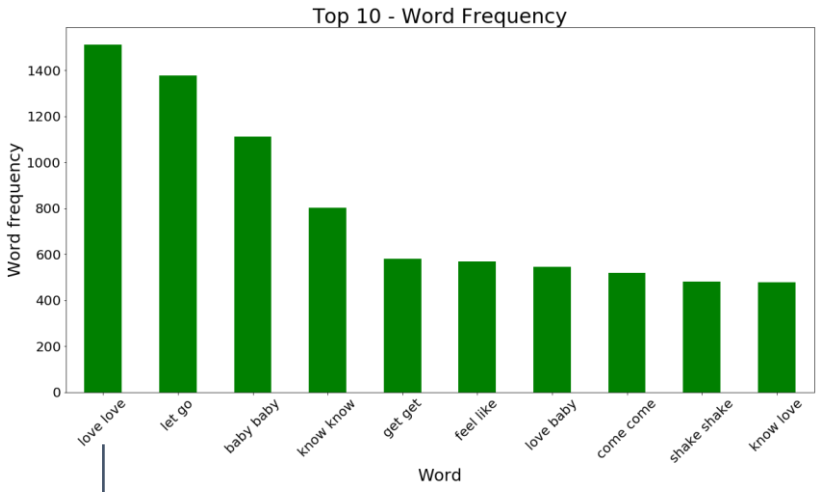
➢ Part I: N-grams -- Most frequent set of words that occur next to each other

Unigram



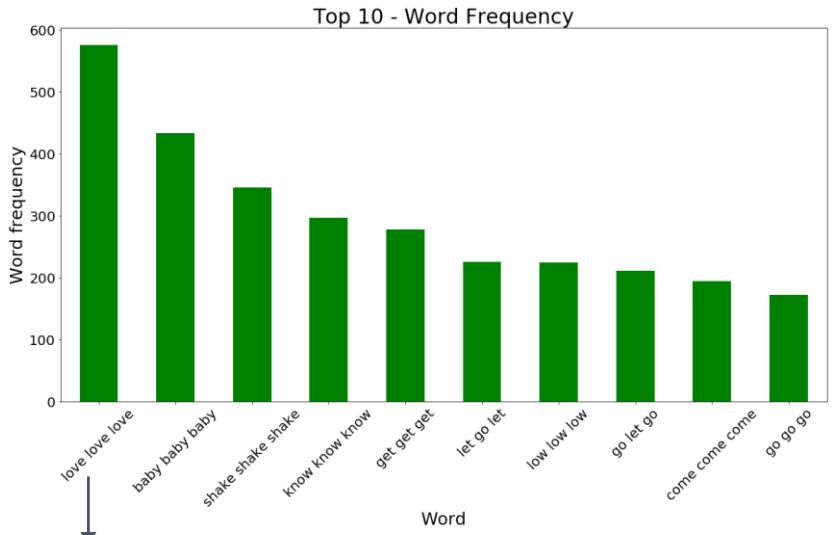
love

Bigram



love love

Trigram

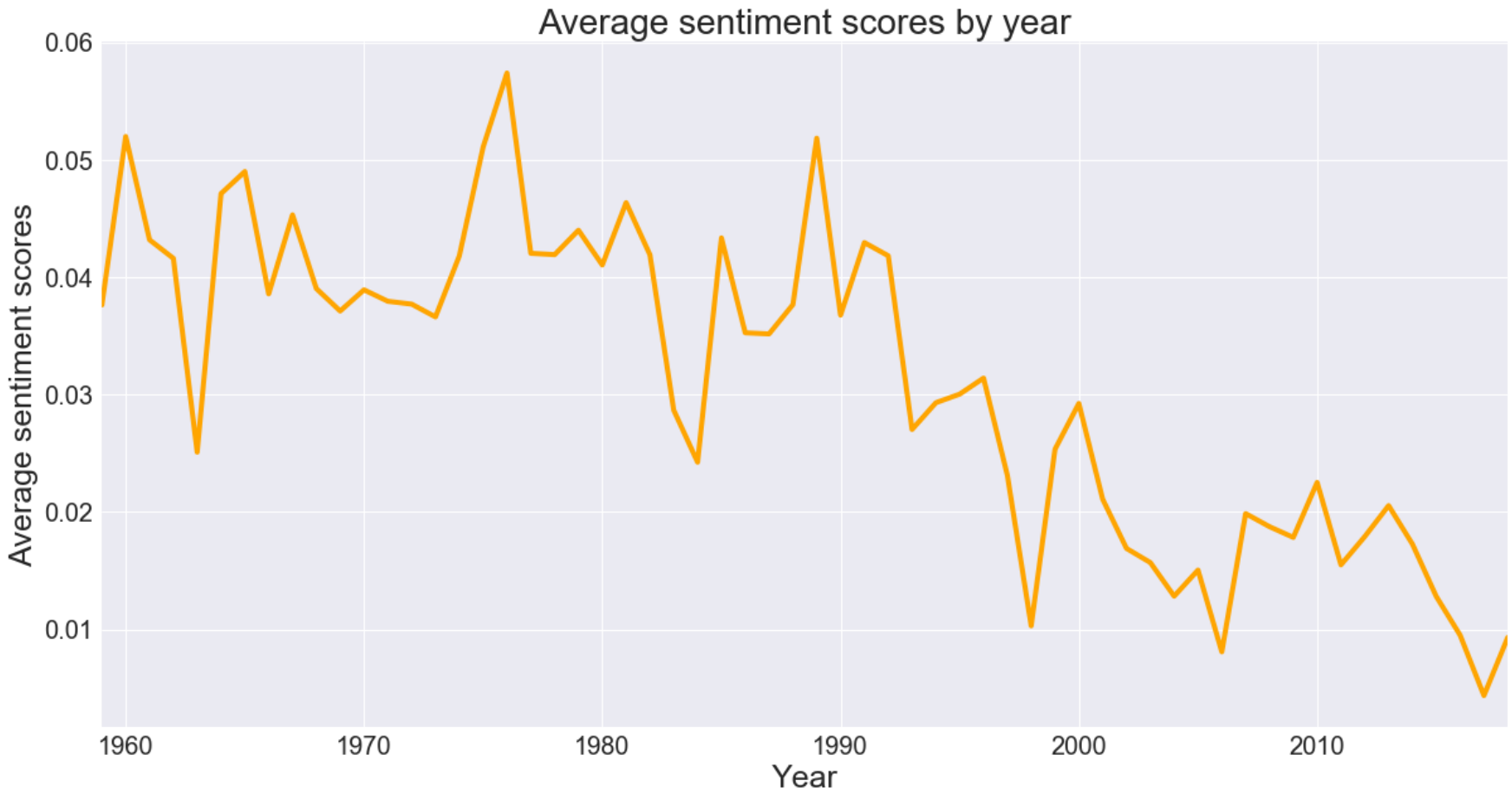


love love love

Love is the way forward!

Text Mining

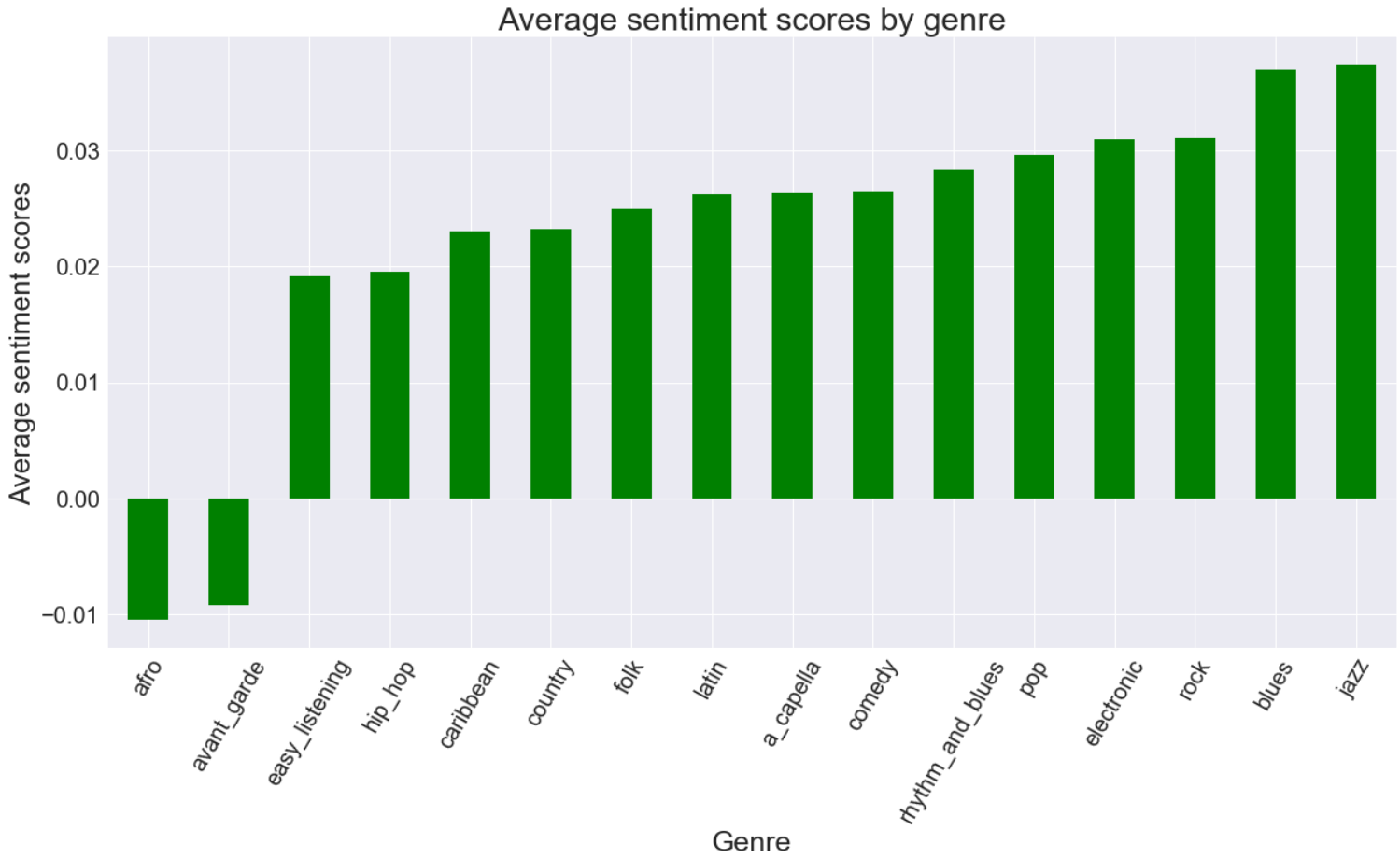
➢ Part II: Sentiment Analysis - Sentiment Intensity Analyzer library



Negative sentiments creeping in!

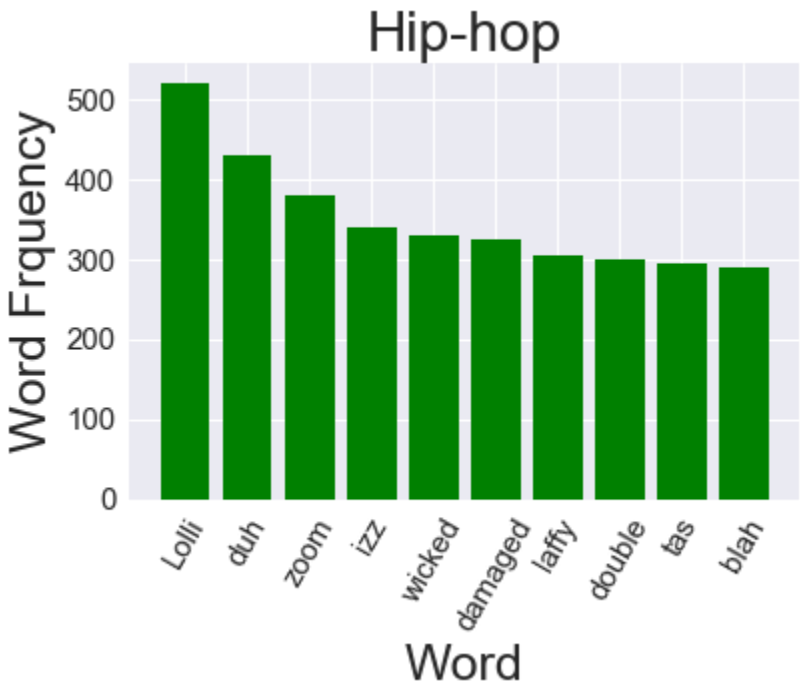
Text Mining

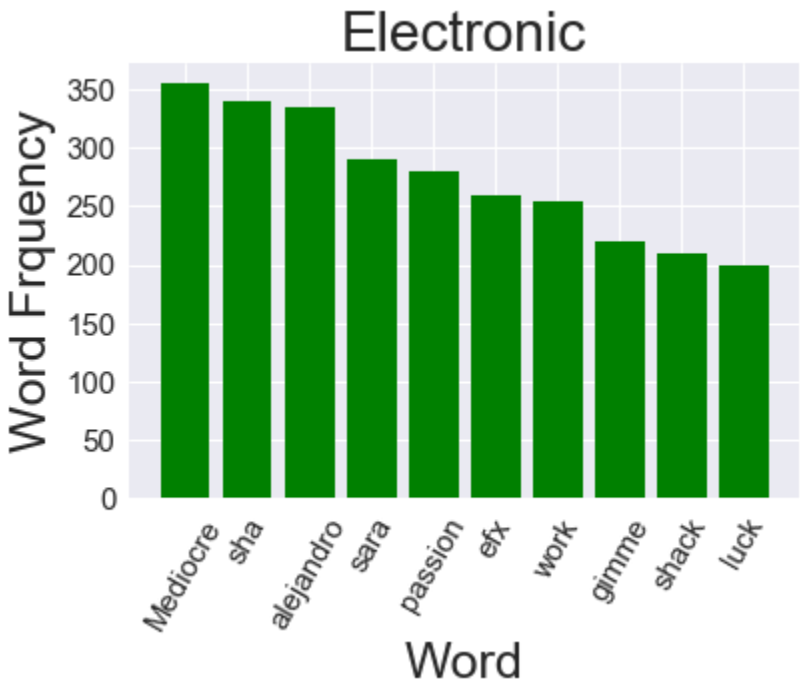
➢ Part II: Sentiment Analysis

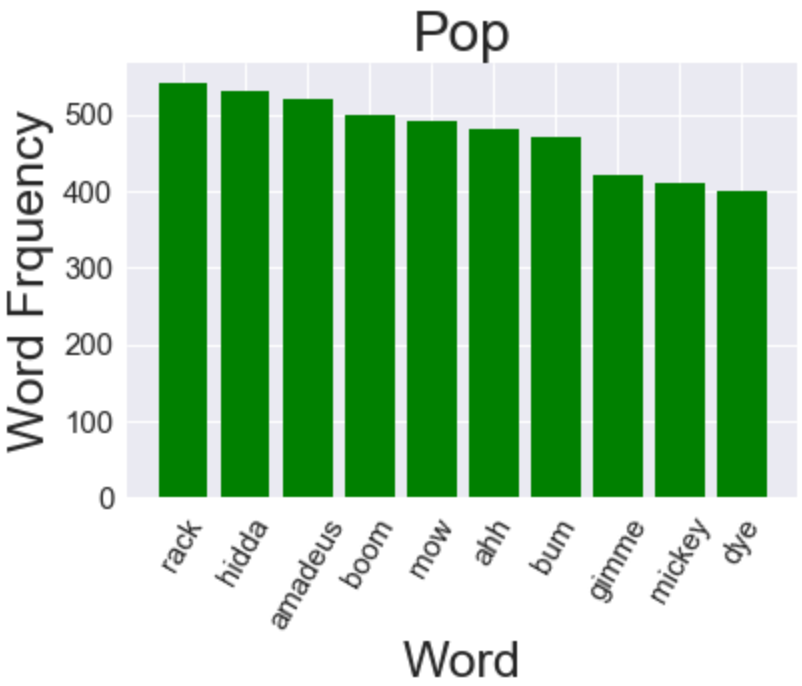


Text Mining

➢ Part III: TF-IDF - Top words encountered for top-3 genres







Hip-hop and Pop have more colloquial word usage!

Conclusion

- Data gathered about Top 100 Billboard songs from 1959-2018
- Data Cleaning for Lyrics, Nationality, Genre of song
- Text Mining - N-gram, Sentiment Analysis, TF-IDF
- More Text Mining - Word Cloud, Parts of Speech Analysis

THANK YOU FOR LISTENING!

Any questions?